

TrackMind AI — Full Prototype Technical Specification

AI-Powered Automatic Block Planning & Disconnection Management System (BDMS) for Indian Railways

Ministry of Railways • Smart India Hackathon • PS 26027 | Full Architectural & Operational Specification

1. Executive Summary & Problem Statement

Indian Railways operates one of the densest and most complex mixed-traffic rail networks globally. Maintenance disconnections ('blocks') are independently requisitioned across three core infrastructure departments: **Engineering (Track / P-Way / TMS)**, **Signal & Telecom (S&T; / SMMS)**, and **Traction Distribution (TRD / TDMS)**. Operating in departmental silos via the Block Demand Management System (BDMS) produces critical operational bottlenecks:

- **Departmental Conflicts:** Independent block requests on identical corridor sections create redundant shutdowns and conflicting track possessions.
- **Severe Timetable Disruption:** Uncoordinated maintenance blocks collide with high-priority express passenger trains and freight rakes, causing cascading delay hours.
- **Asset Underutilization:** Specialized heavy track machinery (tie tampers, ballast cleaners, tower wagons) sits idle awaiting manual safety clearance.
- **Reactive Maintenance:** Disconnections are managed reactively without predictive ML failure probability scoring or automated candidate grouping.

The TrackMind AI Solution: TrackMind AI introduces a fully automated, mathematical and agentic AI block planning engine. It ingests public field defect reports, departmental engineering assessments, corridor availability (COA), live train timetables, and goods traffic forecasts. Using machine learning priority ranking, spatial anomaly detection, constraint-driven candidate classification, and a Google OR-Tools CP-SAT solver, TrackMind AI schedules maintenance blocks with **zero departmental collisions** and enforced timetable constraints while boosting corridor throughput by up to 35%.

2. Complete Technology Stack & Architecture

The system is engineered using modular, high-performance open-source tools selected for reliability and zero-latency execution:

Component Layer	Technologies & Libraries Used	Key Purpose & Functionality
Frontend UI Framework	Streamlit (1.38+), Custom CSS3, Plotly Express	Multi-role reactive dashboard (Central Controller, Engineering, S&T;, TRD) with wide 1160px card containers and Plotly Gantt timelines.
Geospatial GIS Engine	Folium (0.20+), Leaflet.js, B branca	Multi-division interactive map with satellite tiles, station pill markers (DivIcon), animated train vector paths, and speed restriction overlays.
Backend & Database	Python 3.13, SQLite3 (WAL Mode)	Relational database with Write-Ahead Logging and 30s busy timeout for concurrent multi-department operations.
Optimization Engine	Google OR-Tools (CP-SAT Solver)	Constraint programming solver enforcing zero collisions, slot uniqueness, duration feasibility, and timetable protection.
Machine Learning Layer	Scikit-Learn (RandomForest, IsolationForest), Pandas, NumPy	Predicts in-service defect failure probability and identifies spatial defect clustering across 40 section IDs.
Agentic AI Architecture	Plain-Python OOP Framework (32 Agents)	Decoupled autonomous agents handling SLA compliance, cost optimization, crew rosters, TSR lifecycle, and passenger advisories.

Conversational AI & Voice	Groq Cloud API (llama-3.3-70b), Web Speech API	Multilingual voice assistant (English, Hindi, Telugu), multi-intent entity parser, and live Text-to-SQL querying against railway.db.
Reporting Engine	ReportLab (4.2+), FPDF2 (2.8+)	Automated server-side generation of official downloadable block summary reports, audit trails, and technical documentation.

3. Data Architecture & Relational Schemas

The system simulates realistic Indian Railways operations across **5 Divisions** (Secunderabad, Vijayawada, Guntakal, Guntur, Hyderabad) containing **40 Section IDs**. Data is persisted in an indexed SQLite database (`railway.db`) with full transactional integrity.

Database Table	Key Columns / Attributes	Domain Purpose & Workflow Role
defect_intake_v2	defect_id, reported_by_name, role_designation, contact_phone, department, section_id, line_type, nearest_pole_km, description, media_path, status, triage_severity	Public defect submission registry. Stores field observations (restricted strictly to Engineering, S&T,, TRD) awaiting department assessment.
block_requests_v2	request_id, defect_id, department, section_id, track_designation, requested_date, duration_minutes, priority, block_type, description, status	Official engineering block requisitions generated after department assessment with status SUBMITTED / SCHEDULED / REJECTED.
block_candidates	candidate_id, candidate_name, section_id, group_type (ISOLATION, PARALLEL, SEQUENTIAL, REVIEW), total_duration_minutes, request_ids, is_classified, classification_run_id	Preprocessed requisition groups classified by Section Controller engine for conflict-free multi-department batching.
corridor_slots	slot_id, section_id, date, start_time, end_time, duration_hours, slot_type, is_available	Corridor maintenance windows granted by Control Office (Day block, Night block, Maintenance window).
train_timetable	train_no, train_name, train_type, section_id, arrival_time, departure_time, frequency	Official passenger train movement timeline used as a hard constraint to prevent block collisions.
block_allocations / schedule	allocation_id, candidate_id, slot_id, section_id, start_time, end_time, horizon, status, decided_by	Final optimized block allocations solved by CP-SAT solver and confirmed by Section Controllers.
notifications & audit_log	notification_id, recipient_role, message, category, is_read, actor, action, details, timestamp	Full audit trail and 5-line alert notifications dispatched to responsible department heads and controllers.

4. Public Defect Intake & Department Assessment Workflow

TrackMind AI strictly partitions public observation reporting from official engineering diagnosis:

- **Public 'Add Defect' Entry Point:** Accessible via top-right navigation on the login screen. Renders a wide, 1160px horizontal professional form for Loco Pilots, Patrolmen, and station staff. Defect Category dropdown is strictly restricted to the 3 operating departments: **Engineering (Track)**, **Signal & Telecom (S&T;)**, and **Traction Distribution (TRD)**. Public users provide location, line type, nearest pole KM, and observations without setting technical severity (defaults to 'Not Yet Assessed').
- **Automated Department Alerting:** Submissions generate a unique ID (`DEF-YYYYMMDD-XXXX`) and immediately post a standardized 5-line alert to the responsible department queue.
- **AI Advisory Severity Suggestion:** When department engineers open a reported defect, an AI analysis engine inspects description keywords, past fault history, and track class to suggest an advisory severity with detailed rationale (e.g. 'Critical: Rail fracture risks derailment; immediate 30 km/h caution order advised') accompanied by a one-click '[Accept AI Suggestion]' button.
- **Department Engineering Assessment & Requisition:** The authorized engineer confirms official Severity (Low/Medium/High/Critical), enters recommended field actions, and creates a formal block requisition transmitting to `block\_requests\_v2` with status `SUBMITTED`.

5. Candidate Group Preprocessing & Classification Archetypes

When Section Controllers navigate to Candidate Selection & Optimization (``app/controller_requests.py``), the classification engine (``app/classification_engine.py``) automatically evaluates unprocessed requisitions and groups them into 4 operational archetypes:

Classification Archetype	Operational Trigger & Grouping Logic	Scheduling & Safety Impact
ISOLATION	Traction (TRD) power shutdown requisitions requiring 25kV OHE isolation and earthing.	Enforces strict track power-cut windows before field personnel enter high-voltage zones.
PARALLEL (Shadow Blocks)	Multiple department tasks on identical section with compatible durations (e.g. Track renewal + S&T; cable repair).	Merges multiple disconnections into a single unified window, boosting corridor capacity by up to 35%.
SEQUENTIAL	Chained tasks with predecessor/successor safety constraints (e.g. Tie tamping followed by Point machine recalibration).	Enforces strict chronological execution order without allowing overlapping interventions.
REVIEW	Requisitions with conflicting durations, missing resources, or requiring trunk line diversions.	Flagged for Section Controller manual intervention and senior divisional review.

Classification Persistence: All classification states (``is_classified``, ``classification_run_id``, ``classified_at``) are saved directly in SQLite, guaranteeing that candidate counts and categorization remain 100% stable across UI page refreshes and multi-step workflows.

6. Mathematical Optimization Engine (``scripts/optimizer.py``)

The block allocation core solves the combinatorial scheduling problem using **Google OR-Tools CP-SAT** (Constraint Programming - Satisfiability) across weekly (7-day) and monthly (30-day) planning horizons:

Hard Constraints Enforced:

- 1. **Slot Exclusivity:** $\sum_i x_{i,j} \leq 1$ for each corridor slot j (zero departmental collisions).
- 2. **Task Uniqueness:** $\sum_j x_{i,j} \leq 1$ for each defect task i (no redundant allocations).
- 3. **Spatial & Duration Feasibility:** $x_{i,j} = 0$ if `defect.section_id` $\neq$ `slot.section_id` or `defect.duration` $>$ `slot.duration`.
- 4. **Timetable Protection (Hard Constraint):** `filter_slots_against_timetable()` purges any slot whose window overlaps a scheduled passenger train departure.
- 5. **Controller Lock Enforcement:** `filter_slots_against_locked_schedules()` locks manual emergency overrides.

Objective Function: Maximize cumulative priority and failure risk prevention across all scheduled blocks:

Maximize $\sum_{(i,j)} (\text{Priority Score}_i \times 100) \cdot x_{i,j}$

7. Machine Learning Intelligence & 32 Autonomous Agents

Prior to CP-SAT solving, defects are evaluated by a dual ML scoring layer and supervised by 32 decoupled Python domain agents:

- **Multi-Factor Priority Formula:** `Priority Score` = `Severity Weight` (`Critical=40`, `High=25`, `Medium=12`, `Low=5`) + `min(Overdue Days, 90)/90` $\times$ `30` + `min(Trains Affected, 50)/50` $\times$ `30`.
- **RandomForest Failure Risk Classifier:** Predicts probability (0–100%) that a track defect will cause an in-service failure before maintenance.
- **IsolationForest Anomaly Detector:** Evaluates open defect density across 40 sections and flags statistical outliers.
- **32 Domain Agents Framework:** Core Operations (`CoordinatorAgent`, `TrafficAgent`, `ReplanningAgent`), Safety & Governance (`ComplianceAgent`, `SafetyClearanceAgent`, `TSRLifecycleAgent`), Resource Logistics (`CrewAvailabilityAgent`, `TrackMachinePackerAgent`), and Passenger Communication (`PassengerAdvisoryAgent`).

8. User Interface, Multi-Division GIS & ChatMind AI

- **Central Controller & Departmental Dashboards:** Role-based views (Central Controller 10 tabs, Engineering, S&T;, TRD) with secure bcrypt password authentication.
- **Interactive Multi-Division GIS Map:** Built with Folium/Leaflet.js. Visualizes 40 sections across 5 divisions with custom station pill markers (DivIcon), animated train vector paths, track health color coding (Green/Yellow/Red), and TSR overlays.
- **ChatMind AI with Text-to-SQL RAG:** Right-side floating conversational assistant powered by Groq Cloud API ('llama-3.3-70b-versatile'). Features intent parsing for `TRAIN\_SINGLE\_QUERY`, `TRAIN\_DELAYED\_FILTER\_QUERY`, `DB\_QUERY`, `REQUEST\_LOOKUP`, and `DEFECT\_LOOKUP` with direct parameterized SQLite execution for zero-hallucination factual responses.
- **Browser-Native Multilingual Voice AI:** Web Speech API STT/TTS supporting **English** (`en-IN`), **Hindi** (`hi-IN`), and **Telugu** (`te-IN`).

9. Step-by-Step Engineering Build Process & Test Verification

TrackMind AI was engineered across 18 rigorous implementation and validation steps. The full test suite (`scratch/run_all_step_tests.py`) executes **18 test suites containing 241 unit and integration tests** with a 100% pass rate:

Step / Module	Test Suite Name	Tests	Status & Verified Capabilities
Step 1	test_step1_map_data.py	10	PASS - Section coordinates, station pill markers, division bounds.
Step 2	test_step2_train_telemetry.py	10	PASS - Live train vector interpolation, speed calculations, delay HUD.
Step 3	test_step3_block_candidates.py	15	PASS - Requisition intake, candidate table creation, grouping schema.
Step 4	test_step4_corridor_capacity.py	12	PASS - Slot availability, timetable gap identification, slot filtering.
Step 5	test_step5_cp_sat_solver.py	15	PASS - CP-SAT solver, zero collision proof, objective maximization.
Step 6	test_step6_classification.py	15	PASS - Classification engine, ISOLATION/PARALLEL/SEQUENTIAL archetypes.
Step 7	test_step7_block_allocation.py	15	PASS - Allocation engine, schedule persistence, slot locking.
Step 8	test_step8_controller_selection.py	15	PASS - Multi-horizon controller selection, candidate filtering.
Step 9	test_step9_department_notifications.py	15	PASS - 5-line alert formatting, department inbox delivery.
Step 10	test_step10_system_health.py	15	PASS - End-to-end database integrity, WAL concurrency, role access.
Step 11	test_step11_final_acceptance.py	15	PASS - Acceptance criteria, end-to-end workflow validation.
Step 12	test_step12_chatbot_queries.py	15	PASS - Chatbot intent parsing, train queries, SQL generation.
Step 13	test_step13_data_sync.py	12	PASS - Database synchronization, schedule state consistency.
Step 14	test_step14_public_intake.py	15	PASS - Public Add Defect intake, input validation, media storage.
Step 15	test_step15_dept_assessment.py	15	PASS - Department assessment workflow, requisition creation.
Step 16	test_step16_ai_suggestions.py	15	PASS - AI advisory severity suggestion, rationale generation.
Step 17	test_step17_category_restriction.py	12	PASS - Strict 3-department category constraint (Engg, S&T;, TRD).

Step 18	test_step18_classification_persistence.py	10	PASS - Classification state persistence across page refreshes.
---------	---	----	--

End of Technical Specification • TrackMind AI • Smart India Hackathon PS 26027